

Explainability as a non-functional requirement in an AI-assisted veterinary clinical system: a case study

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Abstract

The final abstract must be identical to the abstract included in the final manuscript. It will be inserted after completion and verification of the terminal empirical analysis, since the Results and Conclusions components depend on the experimental evidence.

Keywords

Explainability; Non-functional requirements; AI-assisted diagnosis; Veterinary informatics; User studies

Declarations

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Competing Interests

Pending confirmation before submission. The authors must declare any relevant financial or non-financial interests, including any relevant relationship between the project team and the organization associated with the case study.

Ethics approval and consent to participate

Participation in the requirements-elicitation activities was voluntary and based on written informed consent. Participant information is handled using anonymized identifiers, and identifying materials are maintained separately in the restricted evidence area in accordance with the project's data-protection procedures and Ecuador's *Ley Orgánica de Protección de Datos Personales* (LOPDP).

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Consent for publication

Pending verification before submission. The final statement must reflect the exact scope of the signed consent documents regarding the use and publication of anonymized participant information.

Data, Materials and/or Code availability

The research materials supporting the requirements-engineering phase include anonymized interview transcripts, thematic-coding materials, requirements artefacts, traceability information, and related project documentation. Potentially identifying materials, including original consent documents and multimedia evidence, are maintained separately in the restricted evidence area and are not intended for public dissemination.

The terminal replication package is intended to include the anonymized research materials and reproducible analysis artefacts required by the project. The persistent Zenodo DOI and the corresponding OSF registration identifier will be added once those resources have been finalized. No DOI or OSF identifier is reported until it has been verified.

Author contributions

Pending confirmation by the authors before submission. The final contribution statement will describe the actual contribution of each author, preferably following the CRediT taxonomy.

Use of AI-assisted technologies

During preparation of the manuscript, ChatGPT (OpenAI, GPT-5.6 Luna) was used to assist with drafting, restructuring, language refinement, and consistency checking of manuscript text based on information and research artefacts supplied by the authors.

The AI-assisted tool was not used to fabricate empirical observations, participant responses, statistical results, tables, figures, effect sizes, confidence intervals, or scientific conclusions. Quantitative and qualitative findings included in the final manuscript must be derived from the project's actual evidence and reproducible analysis procedures.

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